

Networking project

Absolutely — let's turn this into a **clean, professional, end-to-end project guide** you can follow without confusion 🙌 🔥

This is exactly how a **real Network Automation Engineer** would structure it.

Project Guide

Python-Driven Network Inventory Auditor with Automated Compliance Validation

OVERVIEW (What You Will Build)

You will build a system that:

- ✓ Connects to network devices
 - ✓ Extracts configurations
 - ✓ Validates security policies
 - ✓ Generates audit reports
-

PHASE 1: Project Initialization

✓ Step 1: Create Project Folder

```
mkdir Network-Automation-Auditor
cd Network-Automation-Auditor
```

✓ Step 2: Create Folder Structure

```
mkdir inventory policies scripts reports output
```

Final Structure:

```
Network-Automation-Auditor/  
|  
├─ inventory/  
├─ policies/  
├─ scripts/  
├─ reports/  
├─ output/  
└─ README.md
```



Guideline:



Always maintain clean structure → **this is what recruiters check**



PHASE 2: Python Environment Setup



Step 3: Create Virtual Environment

```
python -m venv venv
```



Step 4: Activate Environment

```
venv\Scripts\activate
```



Step 5: Install Libraries

```
pip install netmiko pyyaml pandas python-dotenv
```

Guideline:

- Never install globally
 - Always use virtual environment
-

PHASE 3: Configuration Setup

Step 6: Create Inventory File

 inventory/devices.yaml

```
devices:
  - device_type: cisco_ios
    host: 192.168.1.1
    username: admin
    password: admin
```

Purpose:

 Defines **which devices to audit**

Step 7: Create Policy File

 policies/policy.json

```
{
  "ssh_required": true,
  "telnet_allowed": false
}
```

Purpose:

 Defines **security rules**



PHASE 4: Core Script Development



Step 8: Create Main Script



scripts/auditor.py

◆ Step 8.1: Import Libraries

```
from netmiko import ConnectHandler
import yaml
import json
import pandas as pd
```

◆ Step 8.2: Load Files

```
with open('../inventory/devices.yaml') as f:
    devices = yaml.safe_load(f)["devices"]

with open('../policies/policy.json') as f:
    policy = json.load(f)
```

◆ Step 8.3: Connect to Devices

```
connection = ConnectHandler(**device)
```

◆ Step 8.4: Extract Data

```
config = connection.send_command("show running-config")
```

◆ Step 8.5: Compliance Logic

```
ssh_status = "PASS" if "transport input ssh" in config else "FAIL"  
telnet_status = "FAIL" if "transport input telnet" in config else "PASS"
```

◆ Step 8.6: Store Results

```
results.append({  
    "Device": device["host"],  
    "SSH": ssh_status,  
    "Telnet": telnet_status  
})
```

◆ Step 8.7: Generate Report

```
df = pd.DataFrame(results)  
df.to_csv('../reports/audit_report.csv', index=False)
```

Guideline:

👉 Code must be:

- Clean
 - Modular
 - Readable
-

PHASE 5: Execution

Step 9: Run Script

```
cd scripts
python auditor.py
```

✅ Step 10: Verify Output

Check:

```
reports/audit_report.csv
```

🧪 PHASE 6: Testing Without Real Devices

✅ Step 11: Simulate Config

Replace:

```
config = connection.send_command("show running-config")
```

With:

```
config = """
line vty 0 4
  transport input ssh
"""
```

🎯 Purpose:

👉 Test logic without router

📊 PHASE 7: Reporting Enhancement

✓ Step 12: Improve Report

Add:

- Final Status
 - Failure reason
-

Optional:

- Load into Power BI
 - Create dashboard
-

PHASE 8: Documentation (VERY IMPORTANT)

✓ Step 13: Create README.md

Include:

```
# Network Inventory Auditor
```

```
## Features
```

- Automated device login
- Configuration extraction
- Compliance validation
- CSV reporting

```
## Tech Stack
```

- Python
- Netmiko
- Pandas

```
## How to Run
```

1. Activate venv
2. Run script

👉 No README = project is weak ❌

📦 PHASE 9: GitHub Upload

✅ Step 14: Initialize Git

```
git init
git add .
git commit -m "Initial commit"
```

🎯 Add:

- Screenshots
 - Sample report
-

🌟 FINAL OUTPUT

You now have:

- ✓ Automation script
 - ✓ Compliance engine
 - ✓ Structured project
 - ✓ Report system
-

🧠 PROFESSIONAL GUIDELINES (VERY IMPORTANT)

🔥 1. Think Like Engineer, Not Student

Don't say:

❌ "I created a script"

Say:

✅ “Developed an automated network compliance auditing system”

🔥 2. Always Show Impact

- Reduced manual work
 - Improved security visibility
 - Automated auditing
-

🔥 3. Keep It Scalable

Later you can:

- Add multiple devices
 - Add Nornir
 - Add dashboards
-

🚀 NEXT LEVEL (When You're Ready)

I can help you:

- ✅ Add **multi-device automation (Nornir)**
 - ✅ Add **HTML dashboard UI**
 - ✅ Simulate real routers (GNS3)
 - ✅ Prepare **interview explanation (killer)**
-

Just say:

👉 “next level upgrade” 🙌